Content:

Material Characterization

Mechanical Characterisation (Tensile and yield strength, Ductility, Toughness and Hardness), Optical Characterisation, current-Voltage (IV) characterisation, Surface characterisation (Roughness & Crystallinity, particle distribution and magnetic properties.

Instrumentation Techniques:

Principle, construction and working of X-ray Diffractometer, crystallite size determination by Scherrer equation, Principle, construction, working and applications of Atomic Force Microscopy (AFM), X-ray photoelectron spectroscopy (XPS), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Numerical problems.

INTRODUCTION

Knowledge of material and its properties is essential to design and fabricate a product. To fabricate a material, one should know its properties. This can be analysed using various technical instruments which help us to develop a product for specific applications. However, before using the technical instruments, one should acquire the knowledge of its basic principle, working and handling procedure of these instruments. In this chapter we are highlighting the material characterization and instrumentation techniques employed for various engineering applications.

Traditionally materials are classified into three major categories viz., metals, polymers and ceramics. Examples of these are steel, cloth and pottery respectively. Now a day's two more type of materials are

added namely composites and semiconductors based on their applications in aeronautical and semiconductor industry respectively. These materials are used based on their typical physical and chemical properties with respect to specific applications. Some of the typical physical properties like density, melting point, thermal conductivity, electrical conductivity, electrical resistivity, thermal expansion, corrosion resistance etc., are the characteristic features of the material. For example, we can choose Tungsten as a refractory material in Physical Vapour Deposition technique due to its high melting point.

Materials Characterization is an important process in which a material's chemical, microstructure and physical properties are measured using various analytical methods, techniques and tools. It can be classified into two types namely, macroscopic characterization technique and microscopic characterization technique. Macroscopic characterization technique includes tensile and yield strength, ductility, toughness and hardness that can be analyzed using specific instrument like Universal Testing Machine (UTM). However, microscopic characterization includes surface morphology, compositions, grain size, microscopic structure, optical properties etc, can be analyzed using technical instruments like Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM), Atomic Force Microscope (AF), X-ray Diffractometer (XRD) etc., to understand the materials.

MATERIAL CHARACTERIZATION (MACROSCOPIC PROPERTIES).

A Universal testing machine (UTM) is used to test the mechanical properties (tension, compression etc.) of a given test specimen by exerting tensile, compressive or transverse stresses. The machine has been named because of its wide range of tests which can perform over different kind of materials. Various tests like peel test, flexural test, tension test, bend test, friction test, spring test etc., can be performed with the help of UTM.

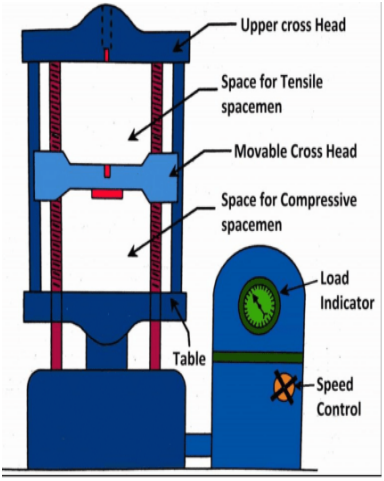


Fig. 1. Component of UTM



Fig. 2 View of UTM

The different components of UTM (both loading and control unit) are labeled and pictorial view of UTM is shown in the Fig. 1 and 2 respectively. UTM has two components namely loading unit and control unit. The arrangement of the test specimen and the exertion of the load are held in the **loading unit**. The variations in the application of the load and the corresponding test result are obtained from the **control unit.**

Construction of UTM

Loading Unit: The loading unit of a UTM consists of three components namely Load frame, Upper crosshead & lower crosshead and elongation scale. The load frame of a UTM can be made either by single support or by double support. The load Frame consists of a table (where the specimen is placed for the compression test), upper crosshead, and lower crosshead. The upper crosshead is used to clamp one end of the test specimen. The lower crosshead in the load frame is the movable crosshead whose screws can be loosened for height adjustment and tightened. Both the crossheads have a tapered slot at the center. This slot has a pair of racked jaws that is intended to grip and hold the tensile test specimen. The relative movement of the lower and upper table is measured by an elongation scale which is provided along with the loading unit.

Control Unit: The main components of the control unit in a universal testing machine are hydraulic power unit, load measuring unit and control devices. Hydraulic power unit consists of an oil pump that provides non-pulsating oil flow into the main cylinder of the load unit. This flow helps in the smooth application of load on the specimen. The oil pump in a hydraulic power unit is run by an electric motor and sump. Load measuring unit has a pendulum dynamometer unit that has a small cylinder with a piston which moves with the non-pulsating oil flow. The pendulum is connected to the piston by pivot lever. The pivot lever deflects based on the load applied to the specimen. This deflection is converted to the load pointer and displays as the load on the dial. The range of load application can be adjusted by means of a knob in the load measuring unit (0-100 kN; 0-250 kN; 0-500 kN and 0-1000 kN). The accuracy of measuring unit controls the overall accuracy of the machine. Control devices can be electric or hydraulic. Electric control devices make use of switches to move the crossheads and switch on/off the unit. A hydraulic control device consists of two

valves, right control valve and left control valve or release valve. A right control valve is used to apply load on the specimen. The left control valve is used to release the load application as shown in the fig.2.

Theory and working. In engineering, tension test is widely used to provide basic design information on the strength of the materials. In the tension test a specimen is subjected to a continually increasing uniaxial tensile force while simultaneous observations are made of the elongation of the specimen. A stress-strain curve is plotted form the load-elongation measurements. The parameters which are used to describe the stress-strain curve of a material are the tensile strength, yield point, percent elongation and reduction of area. The first two gives strength parameters and the last two indicate the ductility of the material.

Working: The main functions of UTM are to test the mechanical properties of materials. The standard tests performed by UTM are: Tensile test, compression test, adhesion tests, pull-out tests, bending test, hysteresis test etc., UTM can be used to test a wide variety of materials like concrete, steel, cables, springs, steel wires and chains, slings, links, rope, winches, steel ropes, etc. based on their application.

UTM provides the value of load application and the respective displacements. From the observed value, the load deflection graph is obtained. With the load value in the Y-axis and the displacement in the X-axis graph will be plotted. The displacement is actually the movement of the crossheads during the load application.

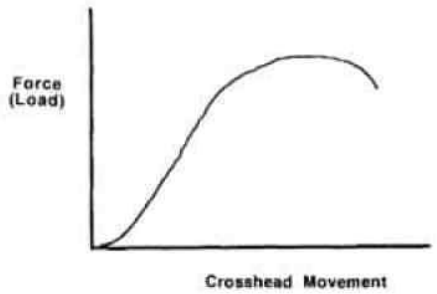


Fig. 3 Force v/s Displacement (cross-head movement) Graph

From the load deflection graph, the stress-strain analysis, modulus of elasticity, yield strength of the specimen tested can be determined.

Analysis: The UTM used for tensile test consists of a hydraulic device for applying a steadily increasing load to a test piece of metal. The load is indicated on a calibrated dial. The material specimen is firmly held in the grips of the testing machine without slippage when the load is applied. It is desirable to test the deformation of the test rod continuously during the tensile test (The specimen is subjected to gradually increasing tensile pull until it breaks). For this purpose the specimen is marked with a 'gauge- length', shorter than the distance between the grips and this will be considered as L_0. This L_0 should be measured every time as and when the reading of the load is taken. The normal stress (load applied)-strain (displacement) curve can be plotted and defined as follows.

$$Stress\ (\tau) = \frac{\text{Load at any instant in N}}{\text{Original cross-sectional area in mm}^2} = \frac{P}{A_o} = \text{N/mm}^2$$

$$Strain(\varepsilon) = \frac{\text{Increase of gauge length}}{\text{original gauge length}} = \frac{L-L_0}{L_0} X100\%$$

When a ductile material is tested under tension, the graph between stress and strain will have a shape as shown in below Fig.4.

Form the figure, the ultimate tensile strength is given by

$$Ultimate\ strength\ (\tau_u) = \frac{Ultimate\ load}{original\ area\ of\ cross-section}$$

$$= \frac{P_u}{P_0}\ N/mm^2$$

Where P_u = maximum load where necking down occurs.

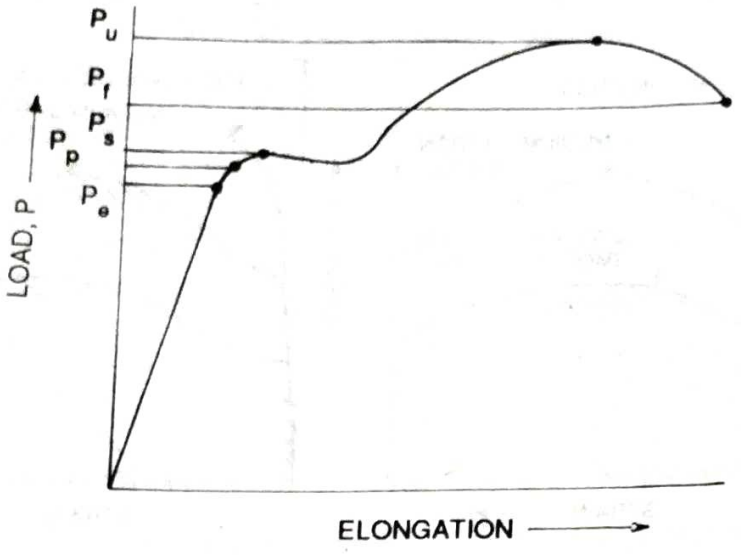


Fig. 4: Stress – strain graph for a ductile material.

When reaching the ultimate tensile strength, test rod is not yet broken, but continuous elongation even at decreasing loads. The ultimate strain is found only at or after rupture (elongation corresponds to P_f in figure) by putting both parts against each other and comparing that length with L_0.

Tensile test for short and long test rods. Consider two test rods of which one is short and the other one is long of same material subjected to tensile test. It is to be noted in the tensile test that the final portion of the elongation is concentrated on a small portion (neck

portion) of the length where the final rupture will arise. The arrangements are shown in fig.5

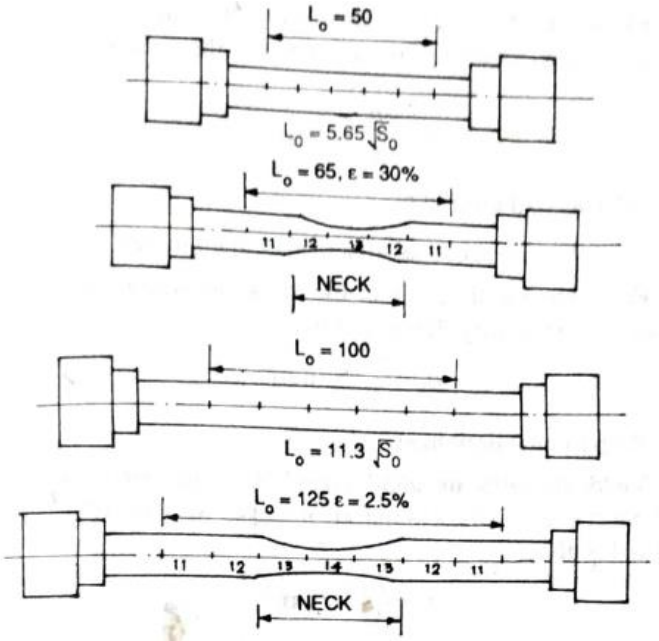


Fig.5: Elongation of short and long test rod under tensile stress.

It can be observed from the figure that the length of a neck portion will be about equal for long and short test rods but the neck elongation will be higher percentage of the ultimate strain in short rod (30%) and a smaller percentage in long rod (2.5%). It means a higher value for ultimate strain in short test rod than in long test rod. To standardize the short and long test rods when length are not normal, ISI has recommended and preferred the length of short test rod as $L_0 = 5.65\sqrt{S_0}$ and for long test rod as $L_0 = 11.3\sqrt{S_0}$ where S_0 is constant.

Other mechanical parameters can be analyzed from Stress- strain graph (refer Fig. 4).

1. Elastic limit: It is the maximum stress that the metal can withstand without experiencing a permanent strain when the load is removed. It is given by

$$Elastic\ limit\ (\tau_e) = \frac{P_e}{A_0}\ N/mm^2$$

Where P_e = elastic load limit in N

A_0 = initial cross sectional area of specimen in mm^2

2. Proportional limit: It is the stress at which the stress strain curve deviates from linearity and is given by

$$Proportional\ limit\ (\tau_p) = \frac{P_p}{A_0} = N/mm^2$$

Where P_p = Proportional limit load.

3. Yield strength or Yield stress: It is the stress at which the material elongates or yields without an increase in load. It is obtained at yield point and is given by

$$Yield\ point\ (\tau_s) = \frac{P_s}{A_0}\ N/mm^2$$

Where P_s = Load at the yield point.

4. Reduction of area. It is the ratio of the decrease in the cross-sectional area of the test piece after fracture of the original area, expressed in percent

$$Reduction\ of\ area(in\ \%) = \frac{d_0^2 - d^2}{d_0^2}\ x\ 100$$

Where d_0= original cross sectional area

d= minimum area of the necked portion.

5. Ductility. Ductility of a material cannot be expressed exactly from tensile test. However, valuable information is obtained from the test curve and by the percentage elongation (i.e., strain) and percentage reduction in area of the test piece.

6. Stiffness. The modulus of elasticity is a measure of stiffness of the material and is given by

$$E = \frac{\tau}{\varepsilon} = \frac{P_e L}{\varepsilon A_0}$$

Where P_p = load at proportional limit

L =gauge length

7. Toughness. Toughness is a combination of strength and plasticity. A tough material can take hard blows without rupturing. Toughness is often defined as a material's ability to absorb energy without cracking. The modulus of toughness is the amount of energy required to cause failure in unit volume of a material and this is represented by the total area under the stress- strain curve. Toughness of some of the materials are shown in below figures (Fig.6).

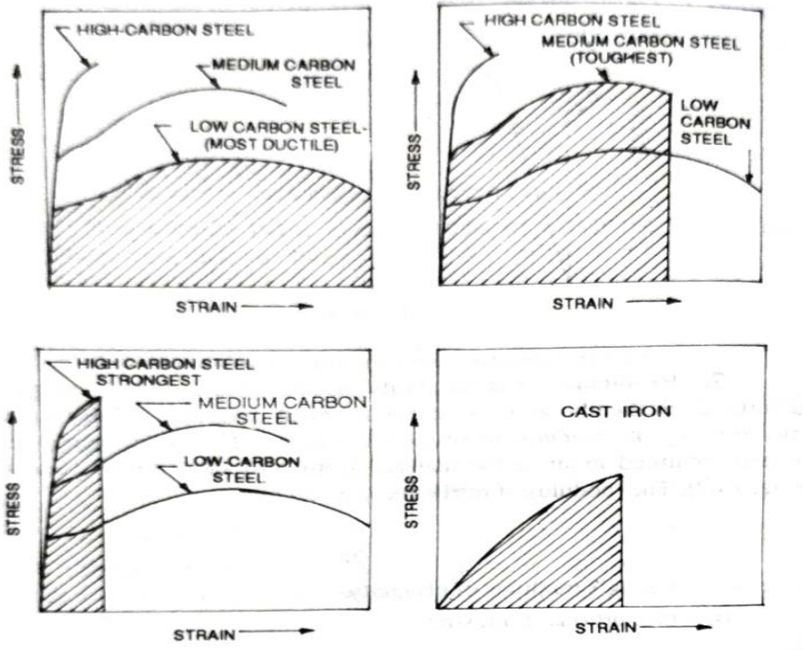


Fig.6 Stress – strain graphs of carbon steel (low, medium and high) and cast iron.

8. Rupture. Rupture is expressed by the breaking strength of the material of the test piece in the tensile test.

Breaking strength = P_f / A_0
Where P_f = load at failure.

Hardness determination:

Hardness is a characteristic of a material, not a fundamental physical property. It is defined as the resistance to indentation, and it is determined by measuring the permanent depth of the indentation. When we apply a fixed force with a given indenter, the smaller the indentation, indicate that the material is harder. Indentation hardness value is obtained by measuring the depth or the area of the indentation. There are various methods to test the hardness of the material.

Different test methods are available based on the type of material and its expected hardness. Materials such as hardened bearing steels have small grain size and can be measured using the Rockwell scale due to the use of diamond indenters and high PSI loading. Materials such as cast iron and powder metals will need a much larger indenter such as used with Brinell scales. Very small parts or small sections may need to be measured on a micro hardness tester using the Vickers or Knoop Scale.

Rockwell Hardness Testing



The Rockwell test is generally easier to perform, and more accurate than other types of hardness testing methods. The Rockwell

test method is used on all metals, except in condition where the test metal structure or surface conditions would introduce too much variation in indentations subjected to sample size and shape.

The Rockwell method measures the permanent depth of indentation produced by a force/load on an indenter. First, a preliminary test force (commonly referred to as preload or minor load) is applied to a sample using a diamond or ball indenter. This preload breaks through the surface to reduce the effects of surface finish. After holding the preliminary test force for a specified dwell time, the baseline depth of indentation is measured.

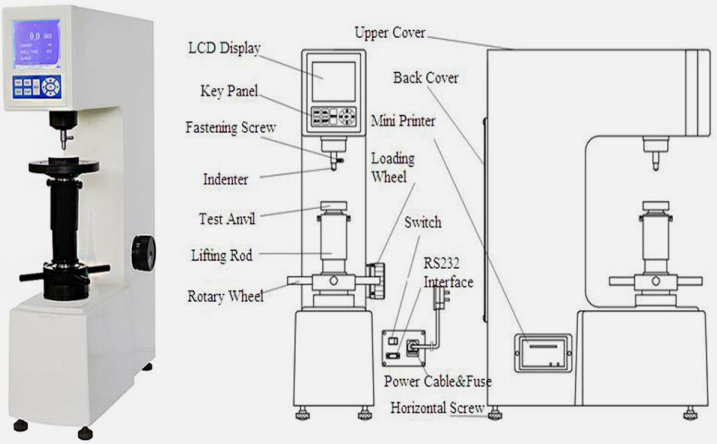


Fig. 7a. View of Rockwell Hardness testing machine and its parts.

After the preload, an additional load, call the major load, is added to reach the total required test load. This force is held for a predetermined amount of time (dwell time) to allow for elastic recovery. This major load is then released, returning to the preliminary load. After holding the preliminary test force for a specified dwell time, the final depth of indentation is measured. The

Rockwell hardness value is derived from the difference in the baseline and final depth measurements and is illustrated in Fig.7b. This distance is converted to a hardness number. The preliminary test force is removed and the indenter is removed from the test specimen.

Preliminary test loads (preloads) range from 3 kgf (used in the "Superficial" Rockwell scale) to 10 kgf (used in the "Regular" Rockwell scale). Total test forces range from 15kgf to 150 kgf (superficial and regular) to 500 to 3000 kgf (macrohardness).

Test Method Illustration

A = Depth reached by indenter after application of preload (minor load)

B = Position of indenter during Total load, Minor plus Major loads

C = Final position reached by indenter after elastic recovery of sample material

D = Distance measurement taken representing difference between preload and major load position. This distance is used to calculate the Rockwell Hardness Number.

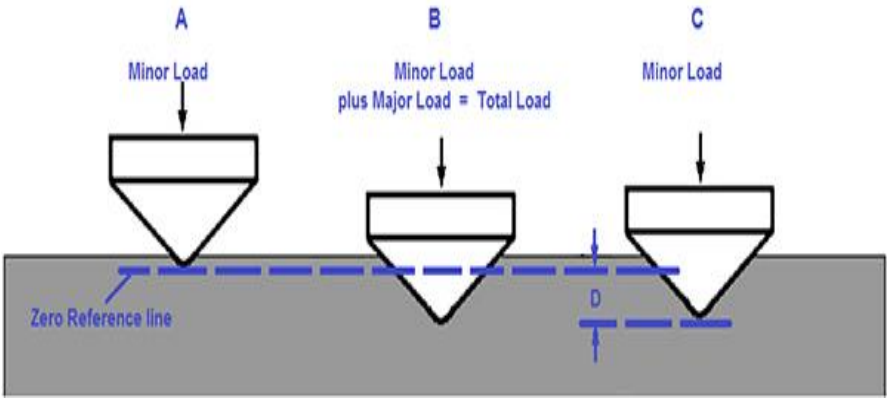
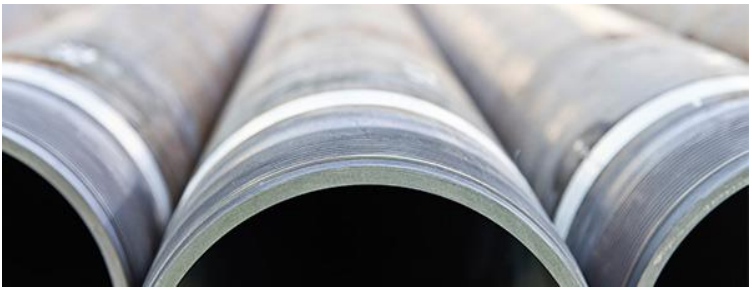


Fig.7b. Illustration of indentation by applying load.

Brinell Hardness Testing



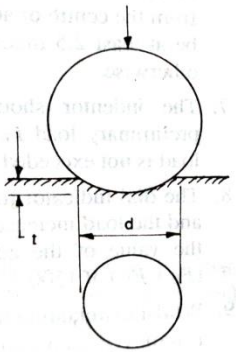
The Brinell hardness test method is used to test materials that have a structure that is too coarse or that have a surface that is too rough to be tested using another test method, e.g., castings and forgings. Brinell testing often use a very high test load (3000 kgf) and a 10mm diameter indenter so that the resulting indentation averages out most surface and sub-surface inconsistencies.

The Brinell method applies a predetermined test load (P) to a carbide ball of fixed diameter (D) which is held for a predetermined time period and then removed. The resulting impression is measured with a specially designed Brinell microscope or optical system across at least two diameters – usually at right angles to each other and these results are averaged (d).

This test consists in forcing a steel (carbide ball) ball of diameter ‘D’ under a load ‘P’ into the test piece and measuring the mean diameter ‘d’ of the indentation left in the surfaces after removal of the load. The Brinell hardness is obtained by dividing the test load ‘P’ by curved surface area of the indentation. This curved surface is assumed to be portion of the sphere of diameter ‘D’. The Brinell Hardness Number is given by

BHN = P/A

Where P = load in N, A = area of contact between the ball and the indentation in mm^2.



If the deformation of the bass during the indentation is neglected, the surface contact is a spherical segment whose area is given by

$A = \pi D t$

Where, D= diameter of the ball, t = depth of the impression.

Let 'd' be the diameter of the impression,

Then, by geometry,

$$t = \frac{(D - \sqrt{D^2 - d^2})}{2}$$

$$BHN = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2}}$$

The Brinell test suffers from some disadvantages:

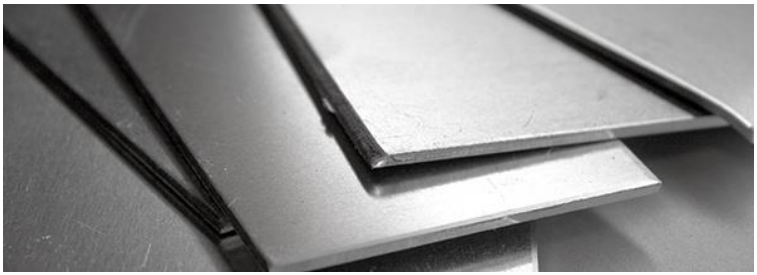
1. It produces a fairly large indentation, particular on soft materials.
2. It cannot be used near the edge of the specimen.
3. It cannot be used on thin metals since reaction from the test anvil might influence test results.

However, irrespective of these limitations, it does give a linear scale of hardness and is particularly useful for design work.

Common test forces range from 500kgf often used for non-ferrous materials to 3000kgf usually used for steels and cast iron. There are

other Brinell scales with load as low as 1kgf and 1mm diameter indenters but these are infrequently used.

Vickers Hardness Testing



The Vickers hardness test method, also referred to as a micro hardness test method, is mostly used for small parts, thin sections, or case depth work. The Vickers method is based on an optical measurement system. The Microhardness test specifies a range of light loads using a diamond indenter to make an indentation which is measured and converted to a hardness value. It is very useful for testing on a wide type of materials, but test samples must be highly polished to enable measuring the size of the impressions. A square base pyramid shaped diamond is used for testing in the Vickers scale. Typically loads are very light, ranging from 10gm to 1kgf, although "Macro" Vickers loads can range up to 30 kg or more. The Microhardness methods are used to test on metals, ceramics, composites - almost any type of material.

The test indentation is very small in a Vickers test, it is useful for a variety of applications: testing very thin materials like foils or measuring the surface of a part, small parts or small areas, measuring individual microstructures, or measuring the depth of case hardening by sectioning a part and making a series of indentations to describe a profile of the change in hardness.

Sectioning is usually necessary with a micro hardness test in order to provide a small enough specimen that can fit into the tester. Additionally, the sample preparation will need to make the specimen's surface smooth to permit a regular indentation shape and good measurement, and to ensure the sample can be held perpendicular to the indenter. The prepared samples are mounted in a plastic medium to facilitate the preparation and testing. The indentations should be as large as possible to maximize the measurement resolution. (Error is magnified as indentation sizes decrease) The test procedure is subject to problems of operator influence on the test results.

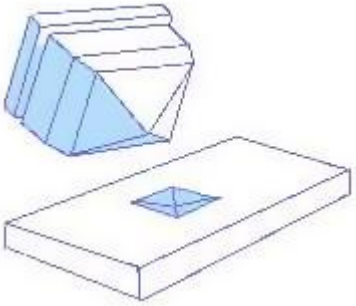


Image:
Opposing indenter faces are set at a 136 degree angle from one another.

Surface characterization:

Magnetic Particle Testing:

Magnetic particle inspection is based on the magnetic lines of flux or force lines that can be generated within the test specimen. These force lines are parallel to each other while travelling through the test specimen if there is no surface or sub-surface defects. The filed lines are shown in the below Figure 8. It can be observed from the top of the figure that the force lines are going through the piece without any deviation. However, if there is a surface defect, the magnetic lines of

force or the flux lines, will create a small north-south pole area at the location of the surface defect (bottom portion of the Fig.8), causing the iron powder that is used in magnetic particle inspection to collect around the surface defect. This surface defects is called a magnetic leakage or an indication. In this technique magnetic field can be produced by strong magnet or an electromagnet (using DC power).

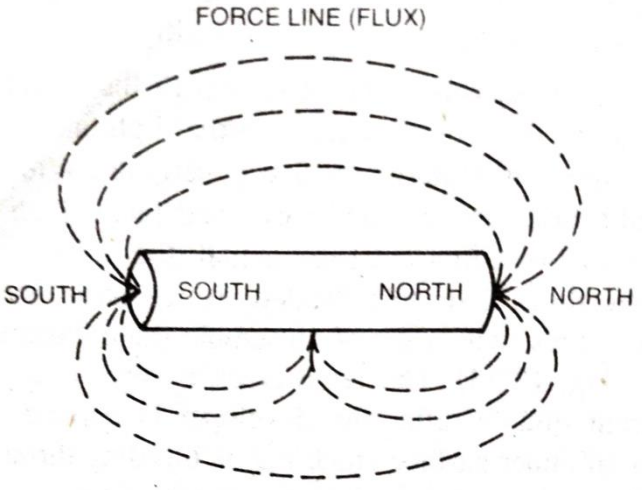


Fig.8: Magnetic lines of force around tiny magnetic particle.

Magnetic particle inspection can also indicate sub-surface defect if these defects are close to the surface. These sub-surface defects disrupt the magnetic lines of force and bend them around the defect. A sub-surface defect is indicated by the magnetic powder collecting around the broken lines of force at the surface of the test specimen. The subsurface defects are shown in the below Fig.9.

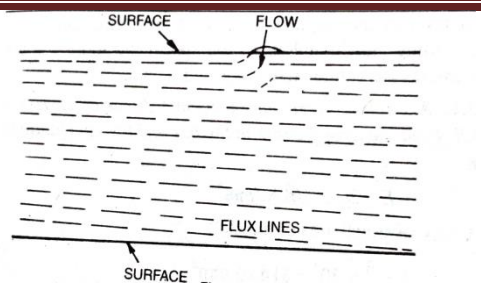
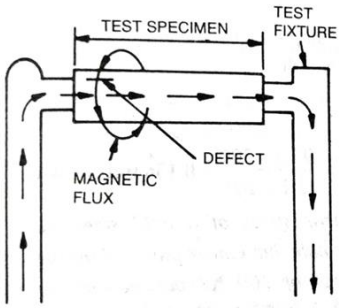


Fig.9 : Subsurface defects indicated by broken lines.

Two types of surface defects can be detected by magnetic particle inspection, defects that are parallel to the interface of the work piece and defects that are perpendicular to the interface or sides, of the test specimen. Longitudinal or parallel defects are indicated because the electric current flow sets up magnetic flux line at a 90^0 angle to the current flow, following the right hand rule of current flow. These magnetic lines of force must cross the defect in a perpendicular line in order to derive the benefit from magnetic particle inspection. If the defect is parallel to the lines of force, there is a possibility that the indication of a surface or a sub surface flow in the test specimen would not be noticeable (See Fig. 10a. Longitudinal defect).

In order to check a test specimen fully, both longitudinal and perpendicular indications must be checked. The perpendicular indications are inspected by using a circular coil which are travelling through the specimen longitudinally. Any defect that is perpendicular to these magnetic lines of flux is shown in below figure (see Fig. 10b. perpendicular defect). Both the longitudinal test and the circular test complete the testing of metal by magnetic particle inspection.



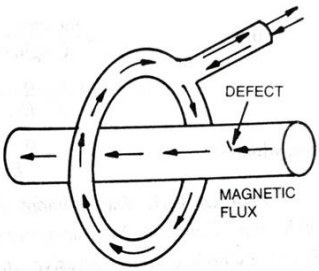


Fig. 10 a): longitudinal effect Fig. 10b): perpendicular defect.

The power requirements for electromagnets are low voltage and high ampere. These are supplied either by direct current or rectified alternating current, both of which will produce a current flowing in only one direction. The particles that are applied to the test specimen can be of different colours according to the colour of the specimen.

MATERIAL CHARACTERIZATION (MICROSCOPIC PROPERTIES)

Microscopic properties refer to properties of atoms while macroscopic properties refer to bulk material. To study the microscopic properties of the material, the material size should be very small of the order of millimeter to micro meter. Material properties like texture, surface morphology, optical and electrical property etc., can be analyzed by applying Physics principle and technical instrument without destroying/spoiling the material (called Non Destructive Technique). Optical and electrical characterization techniques are discussed in the following section.

Optical characterization:

Ellipsometry - *for optical and dielectric characterization*

Ellipsometry is an optical technique for investigating the optical or dielectric properties (complex refractive index or dielectric function and extinction coefficient) and thickness of thin films. The size of the material specimen should be in the form of thin film with the thickness of the order of few mm. Ellipsometry measures the change of polarization upon reflection or transmission and compares it to a model. It is an optical technique used to determine when a parallel beam of monochromatic light of known polarization is incident on a semi transparent thin film, the amplitude of the two perpendicular components of the polarized light and the relative phase difference between them changes due to reflection which results in the film properties like thickness and optical constants, they can be determined using this technique. Actually these parameters cause changes in the ellipticity of the polarized light and hence the name "ellipsometry".

Thus by measuring the ellipsometric parameters viz. the relative change in amplitude (Ψ) and change in phase (Δ) and knowing the refractive index of the material, angle of incidence and wave length of the light, one can calculate the thickness of the film.

Construction:

A typical ellipsometer consists of the following six parts:

- Light source
- polarizer
- photo elastic modulator
- analyzer
- a detector

Light source: It can be either a laser or a lamp (Xenon lamp) emitting continuous spectrum with a monochromator and a focusing system.

Polarizer: It consists of a quartz window and a piezoelectric transducer. It modulates the light beam polarization at 50kHz with the help of computer and control electronics.

The analyzer: It is a linear polarizer which resolves the polarization state of the reflected beam.

The detector: The detector can be any device, able to measure the intensity of the light wave. Generally, two types of detectors are employed viz. photomultipliers and InGaAs photodiodes for UV-vis and NIR applications respectively.

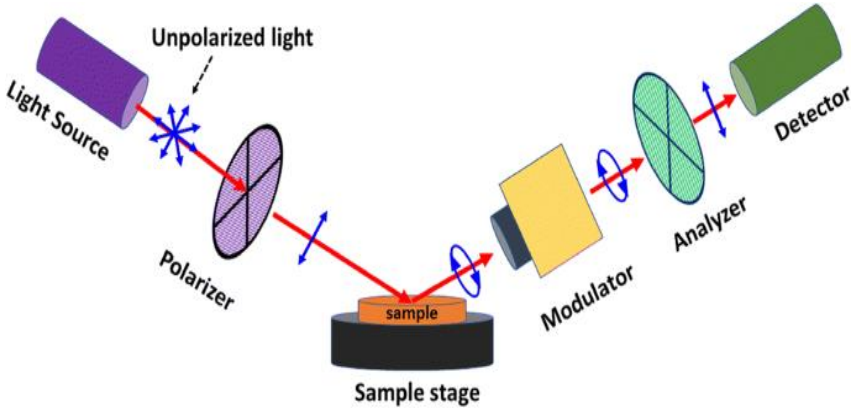


Fig. 11. Schematic Ellipsometric technique.

Working:

A schematic Ellispometric technique is shown in Fig. 11. A light from the source is linearly polarized using a polarizer and is made to be incident on the surface of the sample. The linearly polarized light from the sample surface, becomes elliptically polarized, and travels through a PEM. The function of PEM is to introduce a difference in the phase velocity between the ordinary and extraordinary rays leading to a modulated phase shift. Analyzer examines the incoming lights having different phase velocities and sends the information to the detector. The detector converts the light into respective electronic

signal to determine the polarization of the reflected light. This information is compared with the known input polarization caused by the reflection at the sample.

From the ellipsometric parameters obtained from this experiment one can calculate the refractive index and the thickness of the sample.

For the measurement of complex reflectance ratio ρ of a system, which may be parametrized by the amplitude component Ψ and the phase difference Δ. The polarization state of the light incident upon the sample may be decomposed into an *s* and a *p* component (the *s* component is oscillating perpendicular to the plane of incidence and parallel to the sample surface, and the *p* component is oscillating parallel to the plane of incidence). The amplitudes of the *s* and *p* components, after reflection and normalized to their initial value, are denoted by R^S and R^P respectively. The angle of incidence is chosen close to the Brewster angle of the sample to ensure a maximal difference in R^P and R^S Ellipsometry measures the complex reflectance ratio ρ (a complex quantity), which is the ratio of R^P over R^S

$$\rho = \tan\Psi e^{i\Delta} = \frac{R^p}{R^s}$$

Thus, tanΨ is the modulus of amplitude reflection ratio, and Δ is the phase shift (difference) between s and p polarized reflected light.

Electrical property: Current-Voltage (IV) characterisation (Material: Semiconductor solar cell).

The conducting property (conductivity) of a Semiconducting material can be analysed using current-voltage (I-V) characterization

technique. Let us consider a solar cell of PhotoVoltaic (PV) device madeup of semiconductor photo diode for I-V analysis.

Current –Voltage (IV) characterization of a semiconducting material is the major technique to study the performance of solar efficiency of a solar cell. The **solar Cell I-V characteristic curve** shows the current and voltage (I-V) characteristics of a particular photovoltaic (PV) cell, module or array. It gives a detailed description of its solar energy conversion ability and efficiency. The output performance and efficiency of the device (solar cell) is analyzed using I-V curve.

Introduction: Solar cell is the basic unit of solar energy generation system where electrical energy is extracted directly from light energy without any intermediate process. The working of a solar cell solely depends upon its photovoltaic effect, hence a solar cell also known as photovoltaic cell. A solar cell is basically a semiconductor p-n junction device. It is formed by joining p-type (high concentration of hole or deficiency of electron) and n-type (high concentration of electron) semiconductor material. At the junction, excess electrons from n-type try to diffuse to p-side and vice-versa. Movement of electrons to the p-side exposes positive ion cores in side, while movement of holes to the n-side exposes negative ion cores in the p-side. This results in an electric field at the junction and forming the depletion region. When sunlight falls on the solar cell, photons with energy greater than band gap of the semiconductor are absorbed by the cell and generate electron-hole (e-h) pair. These e-h pairs migrate respectively to n- and p- side of the pn junction due to electrostatic force of the field across the junction. In this way a potential difference is established between two sides of the cell. Typically a solar or photovoltaic cell has negative front contact and positive back contact. A Schematic PV solar cell and its equivalent circuit for I-V measurement is shown in Fig. 12 and Fig. 13 respectively. A semiconductor p-n junction is in the middle of these two contacts like

a battery. If these two sides are connected by an external circuit, current will start flowing from positive to negative terminal of the solar cell. This is basic working principle of a solar cell. For silicon, the band gap at room temperature is Eg = 1.1 eV and the diffusion potential is UD = 0.5 to 0.7 V. Construction of a Si solar cell is depicted in Fig

Solar cell I-V characteristics Curve is the superposition of the I-V curves of the solar cell diode in absence (dark) and in presence of light. Illuminating a cell adds to the normal 'dark' current in the diode so that the diode law becomes;

$$I = I_0 \left[exp\left(\frac{qV}{nkT}\right) - 1 \right] - I_L$$

Where I0 = "dark saturation current"
q = electronic charge
V = applied voltage across the terminals of the diode
n = ideality factor
k = Boltzmann's constant
T = temperature
IL = light generated current

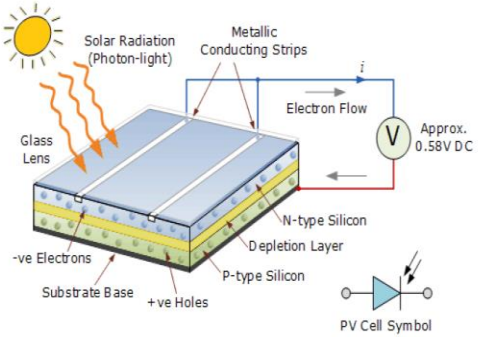


Fig. 12. Schematic of PV solar cell.

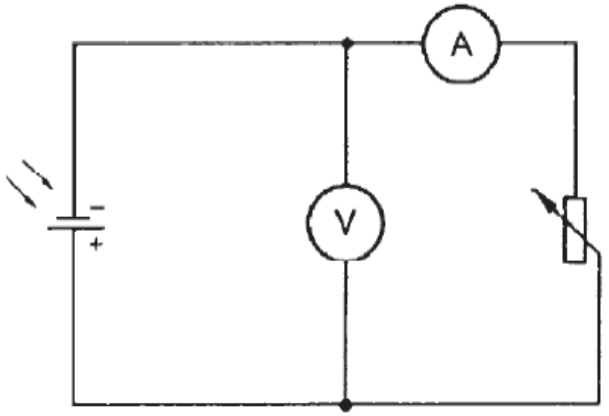


Fig. 13. Equivalent Circuit diagram of I- V measurement of PV solar cell.

A typical circuit for measuring I-V characteristics is shown in Fig.14. From the I-V characteristics various parameters of the solar cell like short-circuit current (Isc), the open-circuit voltage (Voc), the fill factor (FF) and the efficiency can be determined. The rating of solar panel mainly depends on these parameters.

The short-circuit current is the current through the solar cell when the voltage across the solar cell is zero (i.e., when the solar cell is short circuited). This is due to the generation and collection of light generated carriers. For an ideal solar cell at most moderate resistive loss mechanism the short-circuit current and the light- generated current are identical. Therefore, the short-circuit current is the largest current which may be drawn from the solar cell.

The open circuit voltage, Voc, is the maximum voltage available from a solar cell, and this occurs at zero current. The open – circuit voltage corresponds to the amount of forward bias on the solar cell due to the bias of the solar cell junction with the light-generated current.

The 'film factor', more commonly known by its abbreviation "FF" , is a parameter which, in conjunction with Voc and Isc, determines the

maximum power from a solar cell. The FF is defined as the ratio of the maximum power from the solar cell to the product of Voc and Isc. Graphically, the FF is a measure of the "squareness" of the solar cell and is also the area of the largest rectangle which will fit the IV curve as shown in the below figure.

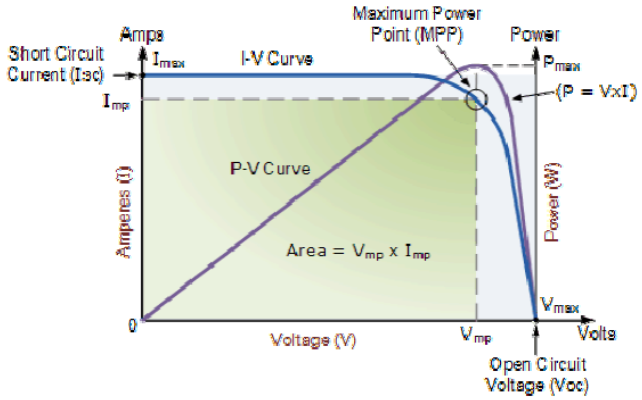


Fig. 14. I-V characteristic of a Solar PV cell.

The efficiency is the most commonly used parameter to compare the performance of one solar cell to another. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. In addition to reflecting the performance of the solar cell itself, the efficiency depends on the spectrum and intensity of the incident sunlight and the temperature of the solar cell.

Surface characterization: (Roughness and Crystallinity): Roughness and crystallinity of the material can be determined by X ray diffractometer and will be discussed along with instrumentation technique.

Instrumentation Techniques:

X-ray Diffractometer

X-ray diffraction (XRD) is a rapid analytical technique primarily used for phase identification of a crystalline material and can provide information on unit cell dimensions. The analyzed material is finely ground, homogenized, and average bulk composition is determined.

Principle: X-ray diffraction is based on constructive interference of monochromatic X-rays and a crystalline sample. These X-rays are generated by a cathode ray tube, filtered to produce monochromatic radiation, collimated to concentrate, and directed towards the sample. The interaction of the incident rays with the sample produces constructive interference (and a diffracted ray) when conditions satisfy Bragg's Law ($n\lambda=2d \sin\theta$). This law relates the wavelength of electromagnetic radiation to the diffraction angle and the lattice spacing in a crystalline sample. These diffracted X-rays are then detected, processed and counted. By scanning the sample through a range of 2θ angles, all possible diffraction directions of the lattice should be attained due to the random orientation of the powdered material. Conversion of the diffraction peaks to d-spacings allows identification of the mineral because each mineral has a set of unique d-spacings. Typically, this is achieved by comparison of d-spacings with standard reference patterns.

Derivation of Bragg's equation

Consider a crystal parallel planes with the inter planar distance 'd' shown in Fig.15a. It can be observed from figure that BB^1, CC1 and DD1 are a set of parallel and equidistance planes of a crystal and the inter-planer distance between these planes are equal to 'd'. Let's consider a beam of monochromatic X-rays PQ is incident on plane

BB^1 with angle θ and reflected along QR. Similarly, another beam XY is incident on the second plane CC^1 with angle θ and reflected among YZ.

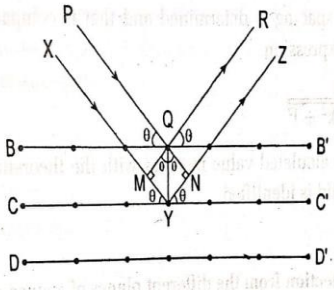


Fig.15a. X ray diffraction on crystal plane.

The path difference between the reflected ray is equal to MY +NY and MY=NY=dsinθ
path difference = MY +NY

= d sinθ + d sinθ

= 2d sinθ, Where d = interplanar distance

Now, in order for reflected rays to be in phase (for constructive interference), the path difference between them must be an integral multiple of the X-ray wavelength (λ), i.e.

nλ = 2d sinθ
This equation is called **Bragg's equation**.

Where n=1,2,3,…..specifies the order of reflection.

[For first-order reflection n=1 and n=2 for second-order reflection and so on]

Construction and working of Diffractometer.

The schematic diagram of Bragg's X-ray diffractometer is as shown in fig 15b.

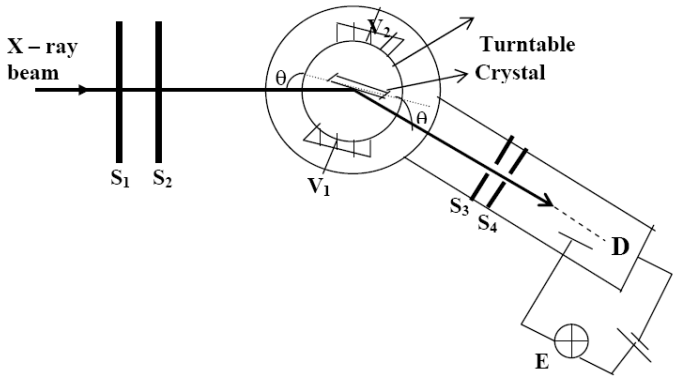


Fig. 15a Bragg's X- ray diffractometer

X –rays produced from an X – ray tube are passed through two lead slits S_1 and S_2 so as to obtain a narrow beam. This fine beam is made to fall on the crystal 'C' fixed on the top of a thin wire mounted exactly at the center of a circular table. The position of the table can be read by means of a vernier scale V_1. The X – ray beam after reflection, enters an ionization chamber D filled with a gas. The ionization chamber is mounted on a mechanical arm which can turn coaxially with the turn table. But the mechanical arm is so coupled to the turn table, for every rotation by an angle θ of the turn table, the mechanical arm turns to an angle 2θ. This is required because, as per the laws of reflection, whenever a reflecting surface turns through an

angle θ, the reflected ray turns through an angle 2θ. Hence the X – rays after reflection from the crystal, are always made to enter the ionization chamber D, and they are again made in to a narrow beam by means of slits S_3 and S_4. The position of the arm carrying the chamber can be read by means of the vernier V_2. The X – rays produce ionization of the gas in the chamber D. The ionization current is measured by the electrometer E. If more X – rays enter the chamber, there will be an increase in the ionization current.

The Bragg's spectrometer is used to

1) Verify Bragg's law, 2) Determine the wavelength of monochromatic x-ray 3) Analysis of crystal structure

Analysis of Crystal structure using Bagg's diffractometer (XRD)



X-ray diffractometers consist of three basic elements: an X-ray tube, a sample holder, and an X-ray detector.

X-rays are generated in a cathode ray tube by heating a filament to produce electrons, accelerating the electrons toward a

target by applying a voltage, and bombarding the target material with electrons. When electrons have sufficient energy to dislodge inner shell electrons of the target material, characteristic X-ray spectra are produced. These spectra consist of several components, the most common being K_α and K_β. K_α consists, in part, of $K_{\alpha1}$ and $K_{\alpha2}$. $K_{\alpha1}$ has a slightly shorter wavelength and twice the intensity as $K_{\alpha2}$. The specific wavelengths are characteristic of the target material (Cu, Fe, Mo, Cr). Filtering, by foils or crystal monochrometers, is required to produce monochromatic X-rays needed for diffraction. $K_{\alpha1}$and $K_{\alpha2}$ are sufficiently close in wavelength such that a weighted average of the two is used. Copper is the most common target material for single-crystal diffraction, with CuK_α radiation = 1.5418Å. These X-rays are collimated and directed onto the sample. As the sample and detector are rotated, the intensity of the reflected X-rays is recorded. When the geometry of the incident X-rays impinging the sample satisfies the Bragg Equation, constructive interference occurs and a peak in intensity occurs. A detector records and processes this X-ray signal and converts the signal to a count rate which is then output to a device such as a printer or computer monitor.

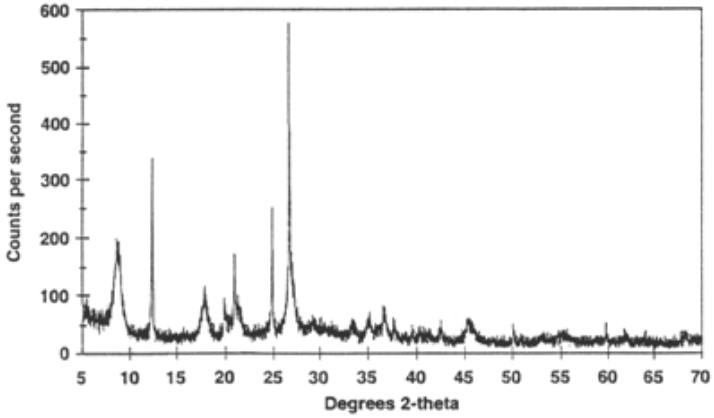


Fig.16. Graph of 2θ Vs counts per second.

The geometry of an X-ray diffractometer is such that the sample rotates in the path of the collimated X-ray beam at an angle θ while the X-ray detector is mounted on an arm to collect the diffracted X-rays and rotates at an angle of 2θ. The graph of 2θ Vs counts are shown in Fig.16. The instrument used to maintain the angle and rotate the sample is termed a *goniometer*. For typical powder patterns, data is collected at 2θ from ~5° to 70°, angles that are preset in the X-ray scan.

Applications

X-ray powder diffraction is most widely used for the identification of unknown crystalline materials (e.g. minerals, inorganic compounds). Determination of unknown solids is critical to studies in geology, environmental science, material science, engineering and biology.

Strength and Limitations of XRD

Strengths

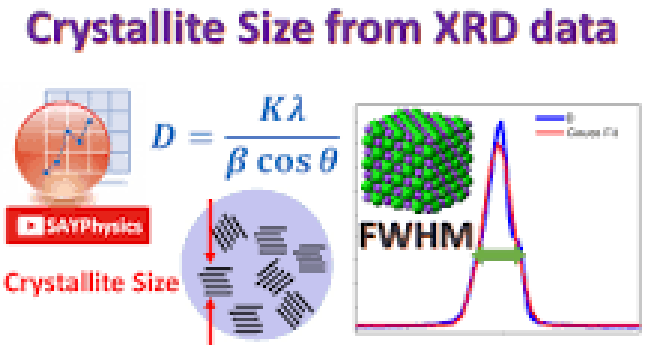
- Powerful and rapid (< 20 min) technique for identification of an unknown mineral
- In most cases, it provides an unambiguous mineral determination
- Minimal sample preparation is required
- XRD units are widely available
- Data interpretation is relatively straight forward

Limitations

- Homogeneous and single phase material is best for identification of an unknown

- Must have access to a standard reference file of inorganic compounds (d-spacings, *hkl*s)
- Requires tenths of a gram of material which must be ground into a powder
- For mixed materials, detection limit is ~ 2% of sample
- For unit cell determinations, indexing of patterns for non-isometric crystal systems is complicated
- Peak overlay may occur and worsens for high angle 'reflections'

Crystallite size determination by Scherrer equation



The crystallite size can be calculated from the full width at half maximum intensity (FWHM) measured on the corrected diffraction profile. The crystal size can be determined by using Scherrer Equation which is given by

$$Scherrer\ Equation\ (D) = \frac{k\lambda}{\beta \cos\theta}$$

Where D is the crystallite size by XRD radiation of wavelength λ (nm) from measuring full width at half maximum of peaks (β) in radian located at any 2θ in the pattern and the Shape factor of K can be 0.62 - 2.08 and is usually taken as about 0.89. The Scherrer

equation, relates the size of sub-micrometre crystallites in a solid to the broadening of a peak in a diffraction pattern.

Validity of Scherrer equation.

The crystallite size can be computed using the Scherrer equation by knowing the full width at half-maximum which is extracted from graph. It is shown that for crystals with linear absorption coefficients below 2117.3 cm(-1) the Scherrer equation is valid for crystallites with sizes up to 600 nm.

X-ray Photoelectron Spectroscopy: (material composition)

X-ray photoelectron spectroscopy (XPS), also known as electron spectroscopy for chemical analysis (ESCA) is primarily used for identifying the chemical species present in the film.

PRINCIPLE: Incident X-rays cause photoemission of electrons from the sample surface. These are energy analyzed, allowing the binding energy of the emitted electron to be obtained.

XPS or ESCA is a surface-sensitive analytical technique that analyzes the energy of the photoelectrons (50-2000 eV) that are emitted when a surface is bombarded with X-rays in Vacuum.

The energy of these electrons is characteristic of the atom being bombarded and thus allows identification of elements [in much the same way as in AES-Auger Electron Spectroscopy].

When X-ray photons are incident on a solid, they eject electrons (photoelectrons) from the valence band. Electrons can be emitted from any orbital, and photoelectron emission occurs as long as the X-ray energy exceeds the binding energy of the electrons. The determination of the kinetic energy of the outgoing electron is the

cornerstone of experimental XPS. The schematic of XPS is shown in Fig.17.

The kinetic energy E_K of the photoelectron is given by $E_K = h\nu - E_B - q\,\phi_{sp}$

where $h\nu$ is the energy of the incident X-ray photon, E_B the binding energy of the electron, and $q\,\phi_{sp}$ is the work function of the spectrometer.

The electron binding energy (E_B) is influenced by its chemical surroundings, and is suitable for determining the chemical states of the atoms by measuring the energy shifts.

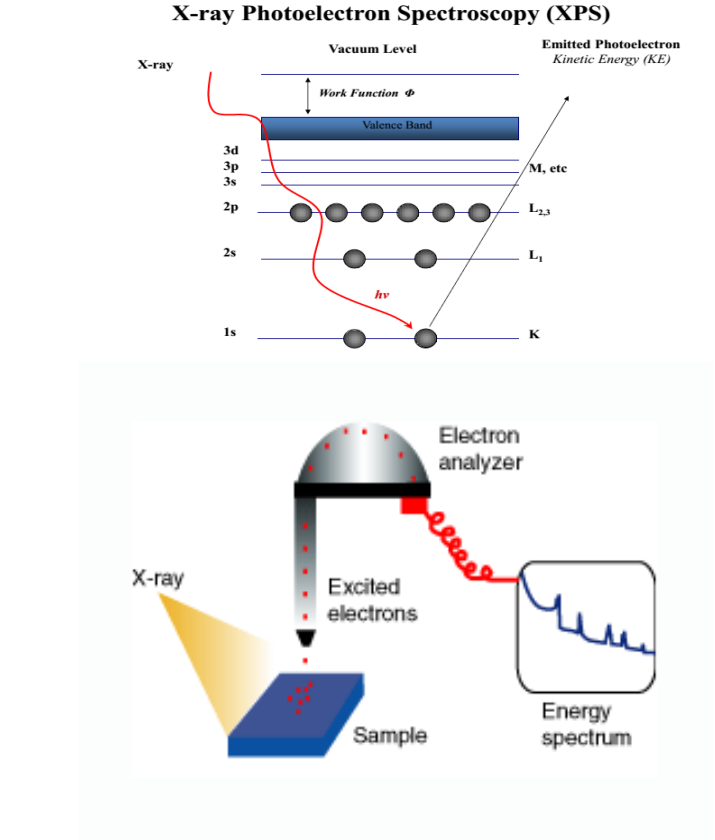


Fig. 17. Schematic of XPS.

Instrumentation:

- ✓ The THREE basic components of XPS are the X-ray source, Spectrometer and a High Vacuum (to protect the sample from contamination).
- ✓ Choice of X-ray source will depend on two factors:
  - X-ray energy must be high enough to excite an intense photoelectron, and
  - X-ray line width should be as narrow as possible; hence typically the K-alpha X-radiation from Magnesium (1253.6 eV) or Aluminum (1486.6 eV) is used for analysis.
  - The XPS electrons are detected with the most common Cylindrical Mirror Analyzer (CMA). A "double-pass CMA" is generally used to discriminate against small energy shifts.
  - Chemical compounds or elements are identified by the location of the energy peaks on the undifferentiated XPS spectrum. Concentration determination is more difficult.
  - "Peak heights and areas" can be used with appropriate correction factors to obtain concentrations, but the method is primarily used for identification.

Advantages & applications:

1) XPS is a "surface sensitive" method because the emitted photoelectrons origibate from the upper 5 – 0.5 nm of the sample.

2) Since X-rays tend to be less destructive when compared with electrons (e -beam), XPS is more suitable for polymers, oxides and organics than AES.

3) XPS/ESCA is primarily used for identifying chemical species at the sample surface. Since the electron binding energy (E_B) is influenced by its chemical surroundings, E_B is suitable for determining chemical states of the atoms by measuring energy shifts.

4) XPS is generally used to determine the ratio of chemical states of compounds in the surface.
5) XPS plays a major role in understanding the chemistry and reaction mechanisms in the development of plasma etching.
6) XPS avoids the electron damage and heating that is sometimes encountered in AES.
7) XPS is one of the primary techniques for analyzing the Elemental, Chemical and Electronic Structure of organic materials. For example, it can determine the chemical environment of each of the carbon atoms in a hydrocarbon polymeric material.
8) Sensitivity of XPS is around 0.1%.

Determination of Surface morphology and topography of the material.

Scanning Electron Microscope (SEM).

The SEM uses electrons instead of light to form an image. SEM is a powerful and commonly used instrument, in both academia and industry for imaging the surfaces of almost any material with a resolution down to about 1 nm. Most of the scanning electron microscopes have magnification ranges from 20X to 100000X. SEM image gives a characteristic three-dimensional appearance and useful for adjudging the surface structure of the sample.

Principle: The smallest distance resolved by a microscope is called Limit of resolution (Δx) and the resolving power of a microscope is the reciprocal of the limit of resolution.

$$\Delta x = \frac{\lambda}{2\mu \sin\theta}$$

$$\text{Resolving Power} = 1/\Delta x = 2\mu\sin\theta/\lambda$$

where $2\mu\sin\theta$ is called the numerical aperture of the objective. Thus shorter the wavelength (λ) of the radiation, greater is the resolving power.

It is this resolving power that sets a limit to the magnification of the microscope, because a mere increase in the magnifying power without a corresponding increase in the resolving power would only produce large blurred image. Optical microscope uses a light beam of wavelength ≈ 600 nm so that the smallest distance resolvable ≈ 300 nm(Assuming numerical aperture = 2). Hence, if instead of light waves we could use electron beam the resolving power can be increased by more than 10^6 fold.

The magnification can be calculated making use of de Broglie waves associated with electrons, accelerated to approximately 75 kV which has a wavelength of 4.48×10^{-12} m which is roughly 10^5 times shorter than visible radiation.

$$\lambda = \frac{h}{\sqrt{2meV}}$$

In optical microscopes, 'light waves' produce the image and magnification is obtained by 'optical lenses'. An electron microscope generally uses electron waves and the electron beam is focused by suitable magnetic lenses.

Magnetic lens:

The magnetic fields which are axially symmetric have a focusing effect on the electron beam passing through them. The axially symmetric magnetic fields are produced by short solenoids. Such solenoids are called

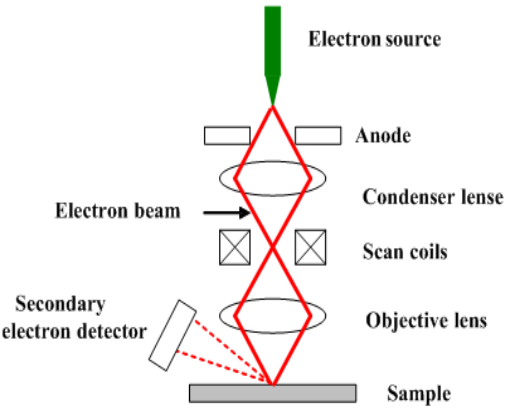


Fig.18. Ray diagram of SEM

magnetic lens. The focal length of such magnetic lens depends on the quality of coil and current. By adjusting the current through the coils the intensity of the magnetic field and the focal length can be adjusted. It is widely used in electron microscopes.

Working Principle of the SEM:

A beam of electrons are produced by heating a thermionic filament. And it is operated with an acceleration voltage of 1 KV to 40 KV for the electrons. The electron beam follows a vertical path and it makes its way through a set of magnetic lenses which focuses and directs the beam towards the sample. These electrons interact with the sample to a depth of ~1μm and generate a number of different types of signals, which are emitted from the area of the specimen where the electron beam is impinging. The induced signals are detected and the intensity of one of the signals (at a time) is amplified and used as the intensity of a pixel on the image on the computer screen. The electron beam, then moves to the next position on the sample and the detected intensity gives the intensity in the second pixel and so on. This produces an image with depth-of-field which is usually 300-600 times better than that of an optical microscope, and also enables a three-dimensional image to be obtained.

Construction:

A schematic ray diagram of SEM is shown in the Fig.18. Electrons are thermionically emitted from a tungsten filament (cathode) are drawn to an anode and focused by two successive magnetic lenses into a beam with a very fine spot size that is typically 10Å in diameter. Pairs of scanning coils located at the objective (magnetic) lens deflect the beam either linearly or in raster fashion over a rectangular area of the specimen surface. Upon impinging on the specimen, the primary electrons decelerate and several processes such as elastic scattering viz., forward scattering and backscattering of

the incoming electrons and inelastic scattering viz., generation of secondary electrons, Auger electrons, Bremsstrahlung, characteristic X-rays, electron-hole pairs (in insulators and semiconductors), long-wavelength electromagnetic radiation etc. A number of different types of signals, which are emitted from the area of the specimen due to the impinging of electron beam is shown in Fig.19.

Various SEM techniques are differentiated on the basis of what is subsequently detected and

imaged, and the principle images produced in the SEM are of three types:

- Secondary electron images,
- Backscattered electron images and
- Elemental X-ray maps.

The secondary electrons are the electrons which are generated from the surface of the specimen when it is bombarded by high energy primary electrons. These electrons are collected by detector which creates a pattern of light and dark areas on a computer screen corresponding to the emission of secondary electrons from the specimen. As the number of electrons produced at any given point can be related directly to the topography of the specimen with respect to the detector, the patterns created on the viewing screen represent the surface topography of the specimen.

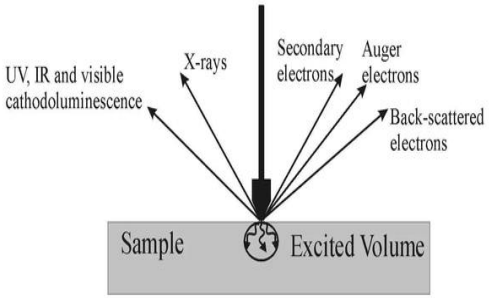


Fig.19. Interaction of electron beam withsample.

Backscattered electrons are the high energy electrons that are elastically scattered and essentially possess the same energy as the incident or primary electrons. The probability of backscattering increases with the atomic number of the sample material. Therefore the primary electrons arriving at a given detector position can be used to yield images containing information on both topology and atomic composition.

An additional electron interaction in the SEM is that the primary electron collides with and ejects a core electron from an atom in the sample. The excited atom will decay to its ground state by emitting either a characteristic X-ray photon or an Auger electron. By analyzing energies of characteristic of the x-ray photon the atoms can be identified. Further the concentration of atoms in the specimen can be determined by counting the number of X rays emitted.

Applications:

The SEM is the most widely employed characterization instrument for obtaining thin film

surface topography and morphology, including

- grain size and shape
- existence of compounds
- formation of textured surfaces
- evidence of film voids
- micro-cracks and lack of adhesion

Atomic Force Microscopy (AFM)

Principle: Atomic Force Microscope (AFM) is a high-resolution microscopy technique which produces precise 3-dimensional topographic images of a sample by scanning the surface with a nanometer-scale probe/tip (lateral resolution ~ 1 nm, vertical ~ 0.1

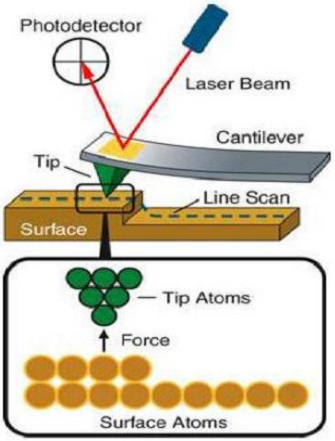
nm) and by measuring attractive or repulsive forces between a tip and the sample.

These deflection are used to study the morphological and textural characteristics like roughness, waviness, lay and flaws of diverse surfaces along with the physical properties like elasticity, adhesion, hardness and friction

Instrumentation: An AFM system consists of a

- Cantilever with a sharp Tip
- Scanner (which moves the tip)
- Microscope (which holds the scanner and sample)
- Controller (which houses all the electronics)
- Computer (used for instrument control and image analysis).

Working: A cantilever with a sharp tip is scanned across the surface of the specimen. The short range or long range forces between the tip of the cantilever and the specimen surface cause the deflection of the cantilever according to Hooke's law. This deflection of the cantilever is measured using the change in the reflection angle of the laser spot incident on the top surface of the cantilever which behaves like a mirror. This reflected laser light is collected by a position sensitive detector to generate a map of the surface topography. Fig.20 represents the schematic of AFM.



There are three main operational modes viz.

i. Contact AFM (Probe-surface separation is < 0.5nm)

ii. Intermittent contact/tapping mode AFM (Probe-surface separation is 0.5-2 nm)

iii. Non-contact AFM (Probe-surface separation range is 0.1 – 10nm)

In contact mode, the cantilever drags across the sample surface and it uses the deflection of the cantilever to measure the contours of the surface. To eliminate the noise and drift that can affect a static signal, low stiffness cantilevers are used, but this allows strong attractive forces to pull the tip to the surface. To eliminate this attraction, the tip is in contact with the surface where the overall force is repulsive.

In non-contact mode, the tip vibrates slightly above its resonance frequency and does not contact the surface of the sample. Any long range forces, like van der Waals forces, decreases the resonant frequency of the cantilever.

In tapping mode, the cantilever uses a piezoelectric element mounted on the top to oscillate it at near to its resonance frequency with amplitude of up to 200nm. The forces cause the amplitude to decrease as the tip gets close to the surface, and the height of the cantilever adjusts to keep the amplitude constant.

This tapping results in less damage to the sample than contact mode and is more accurate than non-contact mode when moisture is present on a sample.

Cantilever and Tip

1) The stiffness of the caintilever needs to be less i.e., the effective spring constant holding atoms together must be in the order of 1 – 10 nN/nm.

2) The tip should have a radius of curvature less than 20-50 nm (smaller the better) with a cone angle between 10-20 degrees.

3) To obtain both low spring constant and high frequency (>2 KHz), mass of the cantilever should be very small (~ 10^{-10} kg).

4) Microlithography process is used to make the cantilevers. Si or Si_3N_4 are used for making tips. A thin gold layer is deposited on upper side of the tip for good reflectivity of laser beam.
5) Tips are generally evaluated by their "end radius" which limits the resolution of AFM.
6) The "normal tip" is a 3 μm tall pyramid with ~30 nm end radius.
7) The electron-beam-deposited (EBD) tip or "supertip" offers a higher aspect ratio (it is long and thin, good for probing pits and crevices) and sometimes a better end radiusthan the normal tip.
8) The "Ultra-lever" is based on an improved microlithography process. Ultra-lever offers a moderately high aspect ratio and on occasion a ~10 nm end radius.

Advantages of AFM

- Easy sample preparation
- Accurate height information
- Works in Vacuum, Air, and Liquids
- Living systems can be studied
- Magnification of the sample along all the three axes.

Limitations of AFM

- Limited vertical range
- Limited magnification range
- Data not independent of tip
- Tip or sample can be damaged

APPLICATIONS of AFM: AFM is a versatile tool to investigate

- Topography of surfaces
- Properties of surfaces – roughness, pin hole formation, defects, grain size analysis, Young's modulus
- Properties of single molecules

- Forces within molecules

Transmission Electron Microscopy (TEM),

Principle: The working principle of the Transmission Electron Microscope (TEM) is similar to the optical microscope. The major difference is that optical microscopes use light rays to focus and produce an image while the TEM uses a beam of electrons to focus on the specimen, to produce an image. Electrons have a shorter wavelength in comparison to light photon which has a long wavelength. The mechanism of an optical microscope is that an increase in resolution power decreases the wavelength of the light, but in the TEM, when the electron illuminates the specimen, the resolution power increases increasing the wavelength of the electron transmission. The wavelength of the electrons is about 0.005nm which is 100,000X shorter than that of light, hence TEM has better resolution than that of the light microscope, of about 1000times. This can accurately be stated that the TEM can be used to detail the internal structures of the smallest particles like a virion particle. The ray diagram of TEM is shown in Fig. 21.

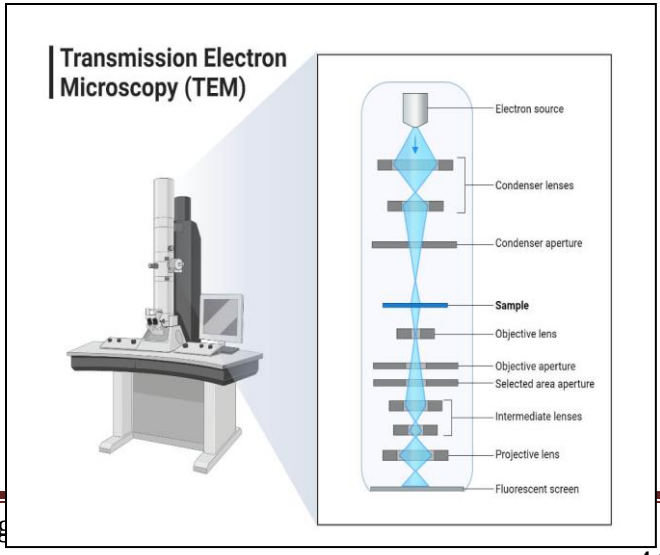


Fig. 21 TEM

Construction of TEM: The three important working parts of TEM are 1) Electron gun 2) Image producing system and 3) Image recording system.

Electron gun

- This is the part of the Transmission Electron Microscope responsible for producing electron beams.
- Electrons are produced by a cathode that is a tungsten filament that is V-shaped and it is normally heated. The tungsten filament is covered by a control grid known as a Wehnelt cylinder made up of a central hole which lies columnar to the tube. The cathode lies on top of or below the cylindrical column hole. The cathode and the control grid are negatively charged with an end of the anode which is disk-shaped that also has an axial hole.
- When electrons are transmitted from the cathode, they pass through the columnar aperture (hole) to the anode at high voltage with constant energy, which is efficient for focusing the specimen to produce an accurately defined image.
- It also has the condenser lens system which works to focus the electron beam on the specimen by controlling the energy intensity and the column hole of the electron gun. The TEM uses two condenser lenses to converge the beam of electrons to the specimen. The two condenser lens each function to produce an image i.e the first lens which has strong magnification, produces a smaller image of the specimen, to the second condenser lens, directing the image to the objectives.

Image- Producing system

- Its made up of the objective lens, a movable stage or holding the specimen, intermediate and projector lenses. They function by focusing the passing electrons through the specimen forming a highly magnified image.
- The objective has a short focal length of about 1-5mm and it produces an intermediate image from the condenser which are transmitted to the projector lenses for magnification.
- The projector lenses are of two types, i.e the intermediate lens which allows great magnification of the image and the projector lens which gives a generally greater magnification over the intermediate lens.
- To produce efficient high standard images, the objectives and the projector lenses need high power supplies with high stability for the highest standard of resolution.

Image-Recording System

- Its made up of the fluorescent screen used to view and to focus on the image. They also have a digital camera that permanently records the images captured after viewing.
- They have a vacuum system that prevents the bombardment or collision of electrons with air molecules disrupting their movement and ability to focus. A vacuumed system facilitates the straight movement of electrons to the image.
- The vacuumed system is made up of a pump, gauge, valves and a power supply.
- The image that is formed is called a monochromatic image, which is greyish or black and white. The image must be visible to the human eye, and therefore, the electrons are

allowed to pass through a fluorescent screen fixed at the base of the microscope.

- The image can also be captured digitally and displayed on a computer and stored in a JPEG or TIFF format. During the storage, the image can be manipulated from its monochromatic state to a colored image depending on the recording apparatus and use of pixel cameras can store the image in color.
- The presence of colored images allows easy visualization, identification, and characterization of the images.

Working of TEM

- A heated tungsten filament in the electron gun produces electrons that get focus on the specimen by the condenser lenses.
- Magnetic lenses are used to focus the beam of electrons of the specimen. By the assistance offered by the column tube of the condenser lens into the vacuum creating a clear image, the vacuum allows electrons to produce a clear image without collision with any air molecules which may deflect them.
- On reaching the specimen, the specimen scatters the electrons focusing them on the magnetic lenses forming a large clear image, and if it passes through a fluorescent screen it forms a polychromatic image.
- The denser the specimen, the more the electrons are scattered forming a darker image because fewer electron reaches the screen for visualization while thinner, more transparent specimens appear brighter.

NOTE: If the screen is moved aside, a photographic image can be captured in pixels forming a permanent image.

Applications of TEM

TEM is used in a wide variety of fields From Biology, Microbiology, Nanotechnology, forensic studies, etc. Some of these applications include:

1. To visualize and study cell structures of bacteria, viruses, and fungi
2. To view bacteria flagella and plasmids
3. To view the shapes and sizes of microbial cell organelles
4. To study and differentiate between plant and animal cells.
5. Its also used in nanotechnology to study nanoparticles such as ZnO nanoparticles
6. It is used to detect and identify fractures, damaged microparticles which further enable repair mechanisms of the particles.

Advantages of Transmission Electron Microscope (TEM)

1. It has a very powerful magnification of about 2 million times that of the Light microscope.
2. It can be used for a variety of applications ranging from basic Biology to Nanotechnology, to education and industrial uses.
3. It can be used to acquire vast information on compounds and their structures.
4. It produces very efficient, high-quality images with high clarity.
5. It can produce permanent images.
6. It is easy to train and use the Transmission Electron Microscope

Limitations of Transmission Electron Microscope (TEM)

1. Generally, the TEMs are very expensive to purchase
2. They are very big to handle.
3. The preparation of specimens to be viewed under the TEM is very tedious.
4. The use of chemical fixations, dehydrators, and embedments can cause the dangers of artifacts.
5. They are laborious to maintain.
6. It requires a constant inflow of voltage to operate.
7. They are extremely sensitive to vibrations and electro-magnetic movements hence they are used in isolated areas, where they are not exposed.
8. It produces monochromatic images, unless they use a fluorescent screen at the end of visualization.

Sample Numericals:

1. A circular steel bar of 20 mm diameter carries a tensile load of 30KN. Find the tensile stress in the bar and the elongation in a length of 300 mm.

Given E = 2.1×10^5 N/mm^2

$$Cross\ ssectional\ area\ of\ the\ bar\ (A) = \frac{\pi}{4} x\ 20^2 = 314.16\ \frac{N}{mm^2}$$

$$Tensile\ stress\ (\tau) = \frac{W}{A} = \frac{30x10^3}{314.16} = 95.5\ N/mm^2$$

$$\frac{Tensile\ stress\ (\tau)}{Tensile\ strain\ (\varepsilon)} = E\ (modulus\ of\ elasticity)$$

$$Tensile\ strain\ (\varepsilon) = \frac{\tau}{E} = \frac{Increase\ in\ length(x)}{Original\ length\ (l)}$$

$$Increase\ in\ length\ (x) = \frac{\tau x l}{E} = \frac{95.5\ x\ 300}{2.1x10^5} = 0.136\ mm.$$

2. The Ultimate shear stress of a mild steel flat is 400 N/mm^2. Calculate the diameter of a hole that can be punched through it without exceeding a compressive stress 200 N/mm^2 in the punch, thickness of the flast being 15 mm. Also, determine the strain in the punch when Young's modulus of its material is 2.1 x10^5 N/mm^2.

Solution: The maximum stress at which a material shears off is known as its 'ultimate shear stress'. Let 'd' mm be the diameter of the hole.

Force required to be applied on the punch for punching the hole = Max. shear stress x area of the plate to be sheared

= 400 x π dt Newton.

But the maximum force that could be applied on the punch = permissible stress x area of the punch

$$= 200 \times \pi d^2/4 \text{ Newton}$$

(here assume the diameter of the punch and the hole to be same)

$$400\ x\ \pi\ dt = 200x\frac{\pi}{4}d^2$$

$$d = \frac{400\ x\ 4x\ dt}{200} = 120\ mm.$$

$$Compressive\ strain = \frac{Compressive\ stress\ in\ the\ punch}{Elastic\ modulus}$$

$$= \frac{200}{2.1\ x10^5} = 0.00095$$

3. Following observations were made during the tensile test of a mild steel specimen:

Diameter of specimen = 40 mm

Length of specimen = 200 mm

Extension under a load of 42.4 kN= $333x10^{-4}$ mm

Load to yield point = 162.1 kN

Maximum load = 252 kN

Length of specimen after fracture = 250.6 mm

Calculate the values of a) Young's modulus b) Yield point stress c) Ultimate stress and d) percentage elongation.

Solution:

a) Load within limit of proportionality W = 42.5 kN and the corresponding extension x = $333x10^{-4}$ mm

$$Crosssectional\ area\ of\ the\ test\ piece\ (A) = \frac{\pi}{4}x40^2$$

$$= 1256.64\ mm^2$$

$$Stress = \frac{W}{A} = \frac{42.4x10^{-3}}{1256.64} = 33.82\ N/mm^2$$

$$Strain = \frac{x}{l} = \frac{333x10^{-4}}{200}$$

$$Young's\ modulus = \frac{Stress}{Strain} = \frac{33.82\ x\ 200}{333\ x\ 10^{-4}} = 2.03x10^5\ N/mm$$

b)

$$Yield - stress = \frac{Load\ to\ yield\ point}{Original\ area\ of\ crossection} = \frac{162.1\ x\ 10^3}{1256.64} = 129\ N/mm^2$$

c)

$$Ultimate\ stress = \frac{Maximum\ load\ applied}{original\ area\ of\ crossection} = \frac{252x\ 10^3}{1256.64} = 200.5\ N/mm^2$$

d)

$$Percentage\ elongation = \frac{(l-L)}{L} x\ 100 = \frac{(250.6-200)x\ 100}{200} = 25.30\%$$

Sample Questions.

1. Explain the construction of Universal Testing Machine (UTM)
2. Explain the working of Universal Testing Machine and analyse the mechanical parameters of a ductile material using stress – strain graph.
3. Explain three types of hardness testing techniques for the material.

4. With a brief procedure explain the determination of surface and subsurface defects by magnetic particle testing.
5. With a neat labelled diagram explain the ellipsometric technique to determine the refractive index and the thickness of the sample.
6. With a neat diagram, explain the current – voltage (I-V) characteristics of photo voltaic solar cell.
7. Derive Bragg's law and explain the construction and working of Bragg's diffractometer.
8.With a neat diagram, explain the working of Xray photo electron spectroscopy (XPS) and give its advantages and applications.
9. With a neat labelled diagram, explain principle, construction and working of Scanning Electron Microscope (SEM).
10. With a neat labelled diagram, explain principle, construction and working of Atomic Force Microscope (AFM).
11. With a neat labelled diagram, explain principle, construction and working of Transmission Electron Microscope (TEM).